PHD WORK PLAN · EHAB HOURANI · USM

Consumer AI Scoring of IGCSE Summaries

Your whole PhD on one page. Follow the stages top to bottom.

65–80	130–160	2	3	2	8
students	scripts	AI tools	passes	human raters	scripts for RQ3

Before	0	**Get ready** Before Sep 2027	→ Team + forms ready
	A	**Write instruments** Sep–Dec 2027	→ Passages, keys, rubric
	B	**Ethics approval** Sep–Dec 2027	→ Approval letter
Year 1 · 2027–28	C	**Pilot + preregister** Jan 2028	→ OSF preregistration
	D	**Students write** Feb–Mar 2028	→ Frozen corpus
	E	**Human marking** Apr–May 2028	→ Locked marks
Year 2 · 2028–29	F	**AI scoring** Winter break 2028–29	→ Session log
	G	**Numbers (RQ1–2)** Jan–Aug 2029	→ Results tables
Year 2 · 2029	H	**AI reasons (RQ3)** Jan–Aug 2029	→ Coded claims
Years 3–4	I	**Papers + thesis** 2029–2031	→ Thesis submitted

Set-up · Data collection · Analysis · Writing

0 Get ready

Before Sep 2027 · You

- [ ] Agree a USM supervisor
- [ ] Ask Rater 2, an anonymiser and a pilot teacher
- [ ] Principal says yes (check if KRG permission is needed)
- [ ] Draft parent consent + student assent forms
- [ ] Install R, RStudio and Atlas.ti (free)

→ You end with: **Team and forms ready**

A Write the instruments

Sep–Dec 2027 · You + 2–3 colleagues

- [ ] Write Passage A, Passage B + a pilot passage (300–400 words, ~12 points + some irrelevant detail)
- [ ] Write a reading key for each passage
- [ ] Write the rubric: Reading /10 (5 levels) · Writing /5 (3 levels)
- [ ] Colleagues rate your passages against 2–3 real past papers
- [ ] Two colleagues check the rubric mapping (full / partial / absent)

✕ Copy Cambridge wording · Put real past papers into ChatGPT/Claude

→ You end with: **Final passages, keys, rubric**

B Get ethics approval

Submit Sep 2027 · approved before Jan 2028 · You + anonymiser

- [ ] Submit the USM ethics application in your first weeks
- [ ] Parent consent + student assent forms (anonymiser collects them)
- [ ] Principal letter · KRG letter if needed
- [ ] Agreements signed by anonymiser and Rater 2
- [ ] Set up encrypted USM storage

✕ Collect any data before approval

→ You end with: **Approval letter**

C Pilot + preregister

Jan 2028 · You, pilot teacher, Rater 2

- [ ] 10–15 students (not yours) write the pilot summary
- [ ] You + Rater 2 mark it · reading ICC below .60? → calibrate again
- [ ] Score with ChatGPT + Claude × 3 passes · time every submission
- [ ] Choose 3 anchor scripts (low, mid, high)
- [ ] Fix the prompt · meet the statistician
- [ ] Preregister the plan on OSF

✕ Change the plan later without a dated note

→ You end with: **OSF preregistration**

D Students write

Feb–Mar 2028 · You (teacher) + anonymiser

- [ ] Anonymiser randomly assigns A→B or B→A
- [ ] Session 1: first passage
- [ ] Session 2: the other passage, on another day
- [ ] Anonymiser codes, removes names, types word for word, re-checks
- [ ] Freeze the typed files · destroy the name list

✕ See names or handwriting · type scripts yourself

→ You end with: **Frozen corpus (130–160 scripts)**

E Human marking

Apr–May 2028 (≥ 4 weeks after writing) · You + Rater 2

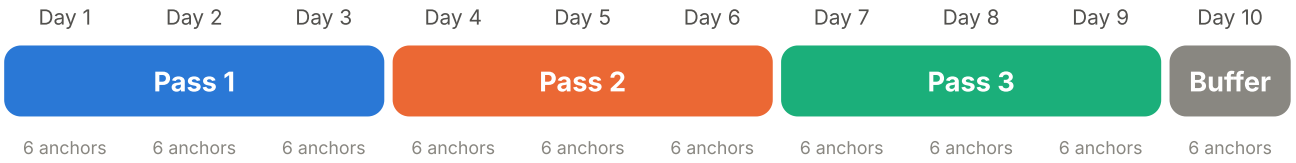
- [ ] Calibrate together on 5–8 practice scripts
- [ ] Mark every script alone, in random order: Reading /10 · Writing /5
- [ ] Totals differ by 3+? Discuss — but keep the original marks
- [ ] Lock the marks file · send a copy to your supervisor
- [ ] Check H1–H2 ICC ≥ .60 (below = report as a limitation)

✕ Look at any AI score · discuss marks while marking

→ You end with: **Locked marks file**

F AI scoring

Winter break, Dec 2028–Jan 2029 · You



- [ ] Same named model every time · memory, training, custom instructions OFF · temporary / incognito chat
- [ ] Each script: new chat → paste prompt → send → copy reply + marks → status → screenshot → close
- [ ] Invalid answer? One retry only, logged
- [ ] Back up the log every evening

✕ Edit the prompt · regenerate answers · finish in another holiday

→ You end with: **Full session log (840–1,020 rows)**

G Numbers (RQ1 + RQ2)

Jan–Aug 2029 · You + statistician

- [ ] **RQ1** same mark twice? → ICC(1,1) across the 3 passes
- [ ] **RQ2** agree with humans? → QWK + ICC(A,1) vs H1 and H2, each pass
- [ ] Correction gap = AI − human average (+ lenient, − harsh)
- [ ] Bland–Altman plots · mixed model (tool × passage + pass)
- [ ] 95% CIs: bootstrap by student, 2,000 times

✕ Report the 3-pass average as the main result

→ You end with: **Results tables + figures**

H AI reasons (RQ3)

Jan–Aug 2029 · You + Rater 2

- [ ] Pick 8 scripts, 4 per passage: biggest reading gap · biggest writing gap · opposite direction · closest agreement
- [ ] Split the 16 AI reasons into numbered claims (~100)
- [ ] Each claim: Supported · Contradicted · Too vague · Ambiguous
- [ ] Both tick codes in Atlas.ti separately
- [ ] Kappa first → one meeting → written log (no recording)

✕ Call a short reason an error · say it happens because students are multilingual

→ You end with: **Coded claims + kappa**

I Papers + thesis

2029–2031 · You + supervisor

- [ ] Paper 1: RQ1 → Language Testing / Language Assessment Quarterly
- [ ] Paper 2: RQ2 + RQ3 → Assessing Writing / Computers & Education: AI
- [ ] Thesis chapters 1–7
- [ ] List every change from the preregistration

✕ Claim teachers can now trust AI marks

→ You end with: **Thesis submitted (Aug 2031)**

Supervisor questions — short answers

What is new in this PhD?
Separate reading and writing marks + checking each AI reason against the text, under real teacher-style access.

How is it different from Sawaki et al. (2025)?
They used checklist items with university students; I use IGCSE reading/writing marks, teenagers, 2 passages, 2 coders.

Why not the API?
Teachers use the apps, not the API. I study what teachers actually get.

Why not free versions?
Out of scope: they can use different models and limits.

Why 3 passes?
Three scores per script show variation better than two, and still fit one holiday.

Why only one prompt?
Keeps the study focused; prompt effects are future research.

Why write your own passages?
No copyright, not in AI training data, full control; checked against real papers.

Your rubric isn't Cambridge's?
It mirrors Tables A and B; two colleagues check every row.

What if the AI model changes?
I pick the same named model each time; if it still changes, the backup rule applies.

Are 65–80 students enough?
Yes, with honest (wide-ish) confidence intervals; precision depends on students, not AI scores.

Why bootstrap by student?
Each student has 2 scripts and many scores; this keeps intervals honest.

Why is reading primary?
It carries 10 of 15 marks and has the longer scale.

What if you and Rater 2 disagree?
Calibrate; if ICC is still below .60, I report it as a limitation.

Are 8 scripts enough for RQ3?
Yes for illustration; I don't claim how often problems happen.

Do AI reasons show how AI really scored?
No. They are what a teacher reads, not the model's inner process.

Does AI disadvantage multilingual students?
I can't say that: there's no comparison group.

You are teacher, researcher and rater?
Anonymiser handles names and typing; blind marking; second rater; 4-week gap.

Is putting student work into AI ethical?
Only coded, typed, anonymous text; training off; temporary chats.

Can you score ~1,000 in two weeks?
Yes: about 5 hours a day for 9 days + a buffer day; the pilot confirms timing.

What if AI agrees closely with humans?
That's a finding too: agreement under real teacher conditions.